Introduction to CUDA — GPU Programming Fundamentals

Device Query, Array Reduction & Matrix Addition

CUDA C (nvcc 13.2) Status

> **UCS645: Parallel & Distributed Computing | Assignment 6 — Introduction to CUDA**

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System Configuration

Component	Details
CPU	Intel Core i7-14700HX (20 cores, 28 threads)
GPU	NVIDIA GeForce RTX 5060 Laptop GPU (Blackwell)
Compute Capability	12.0
SMs	26
VRAM	8 GB GDDR7 (128-bit bus)

Component	Details
Theoretical BW	384.0 GB/s
OS	Fedora 43 (Linux 6.19.12)
CUDA Toolkit	13.2
Compiler	nvcc 13.2 with -O2 -arch=native

Project Structure

```
LAB6/
├── Makefile
├── exercises/
│   ├── a_device_query.cu
│   ├── b_array_sum.cu
│   └── c_matrix_add.cu
├── graphs/
│   ├── b_array_sum.png
│   ├── c_matrix_add.png
│   └── dashboard.png
├── report.md
└── Assignment_6_Report.pdf
```

Part A: Device Query

Problem

Write a CUDA device query program and answer 10 questions about GPU hardware capabilities.

Device Properties

Property	Value
GPU Name	NVIDIA GeForce RTX 5060 Laptop GPU
Architecture	Blackwell (Compute 12.0)

Property	Value
SMs	26
Max Threads/Block	1024
Max Block Dims	(1024, 1024, 64)
Max Grid Dims	(2147483647, 65535, 65535)
Warp Size	32
Shared Memory/Block	48 KB
Shared Memory/SM	100 KB
Global Memory	7706 MB
Constant Memory	64 KB
L2 Cache	32768 KB
Memory Clock	12001 MHz
Memory Bus	128-bit
Double Precision	YES

Answers to Questions

Q1. Architecture is **Blackwell**, compute capability **12.0**.

Q2. Max block dimensions: **(1024, 1024, 64)** with max 1024 threads per block.

Q3. With max grid=65535 and max block=512: **65535 x 512 = 33,553,920 threads**.

Q4. A programmer might not launch max threads when: the problem size is smaller than max threads, register/shared memory pressure limits occupancy, or the algorithm requires specific thread counts (e.g., power-of-2 for reduction).

Q5. Factors limiting max threads: register usage per thread, shared memory per block, warp scheduling limits, problem size, and device memory capacity.

Q6. Shared memory is fast on-chip SRAM accessible by all threads in a block (latency ~20-30 cycles vs ~400 cycles for global). This GPU has **48 KB per block**, **100 KB per SM**.

Q7. Global memory is off-chip DRAM accessible by all threads. This GPU has **7706 MB** of GDDR7.

Q8. Constant memory is read-only cached memory optimized for broadcast (all threads read same address). This GPU has **64 KB**.

Q9. Warp size (32) is the number of threads executing in SIMT lockstep. All threads in a warp execute the same instruction. Divergent branches within a warp serialize execution.

Q10. Yes, double precision is supported (compute capability >= 1.3).

Part B: Array Sum

Problem

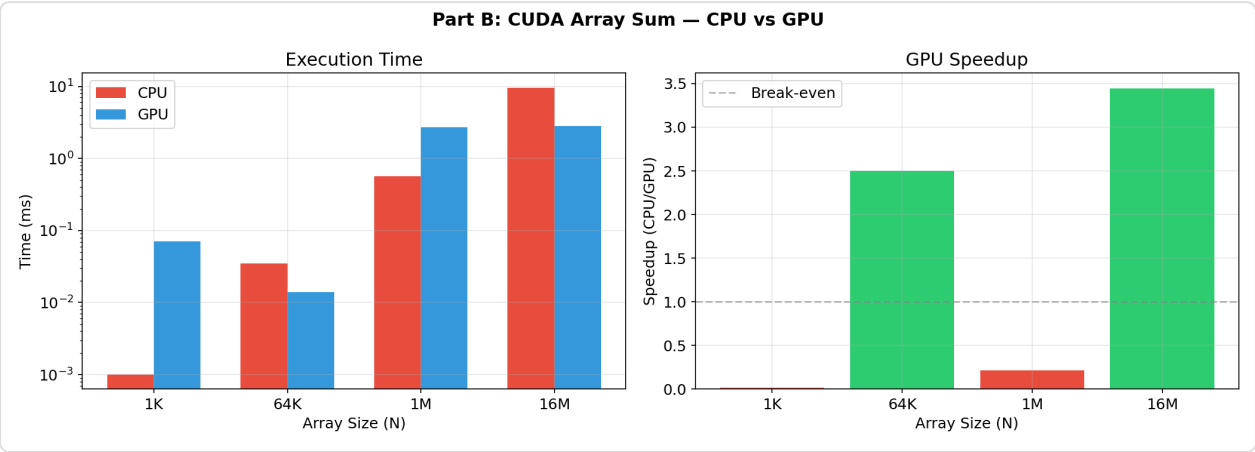
Write a CUDA kernel to compute the sum of a float array using parallel reduction.

Approach

- **Parallel tree reduction** in shared memory: each block reduces its chunk to a partial sum
- Multi-pass: first pass reduces N elements to `ceil(N/256)` partials, subsequent passes reduce until 1 element remains
- `__syncthreads()` ensures correctness between reduction steps
- CUDA Events for GPU timing, `clock_gettime` for CPU baseline

Results

Array Size	CPU Time	GPU Time	Speedup	Verification
1,024	0.001 ms	0.071 ms	0.01x	[PASS]
65,536	0.035 ms	0.014 ms	2.5x	[PASS]
1,048,576	0.567 ms	2.699 ms	0.2x	[PASS]
16,777,216	9.649 ms	2.799 ms	3.4x	[PASS]



Analysis

GPU becomes faster than CPU only at large sizes (16M elements, 3.4x speedup). At small sizes, kernel launch overhead (~5-10 us) and multi-pass reduction memory allocation dominate. The reduction kernel is also not optimized — it uses multiple cudaMalloc calls during the multi-pass phase. A single-pass atomic reduction or warp-shuffle reduction would perform better. The crossover point is between 64K and 16M elements, indicating that PCIe transfer and launch overhead require significant computation to amortize.

Part C: Matrix Addition

Problem

Perform matrix addition of two large integer matrices in CUDA and analyze floating-point operations and memory access patterns.

Approach

- 2D grid/block layout: `dim3(16, 16)` threads per block
- Each thread computes one element: `C[i] = A[i] + B[i]`
- Grid sized to cover full matrix with ceiling division

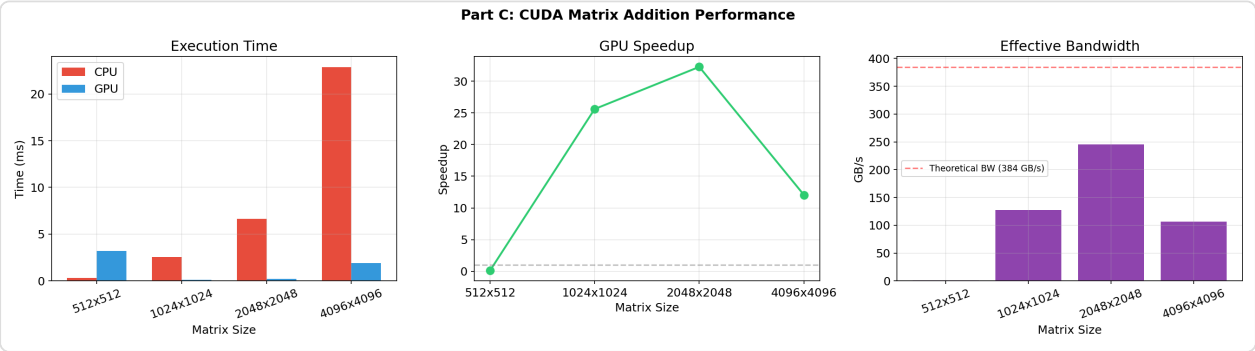
Results

Matrix Size	CPU (ms)	GPU (ms)	Speedup	Eff. BW (GB/s)
512x512	0.325	3.219	0.1x	1.0

Matrix Size	CPU (ms)	GPU (ms)	Speedup	Eff. BW (GB/s)
1024x1024	2.535	0.099	25.7x	127.4
2048x2048	6.614	0.205	32.2x	245.1
4096x4096	22.850	1.896	12.1x	106.2

Memory access analysis for NxN matrix:

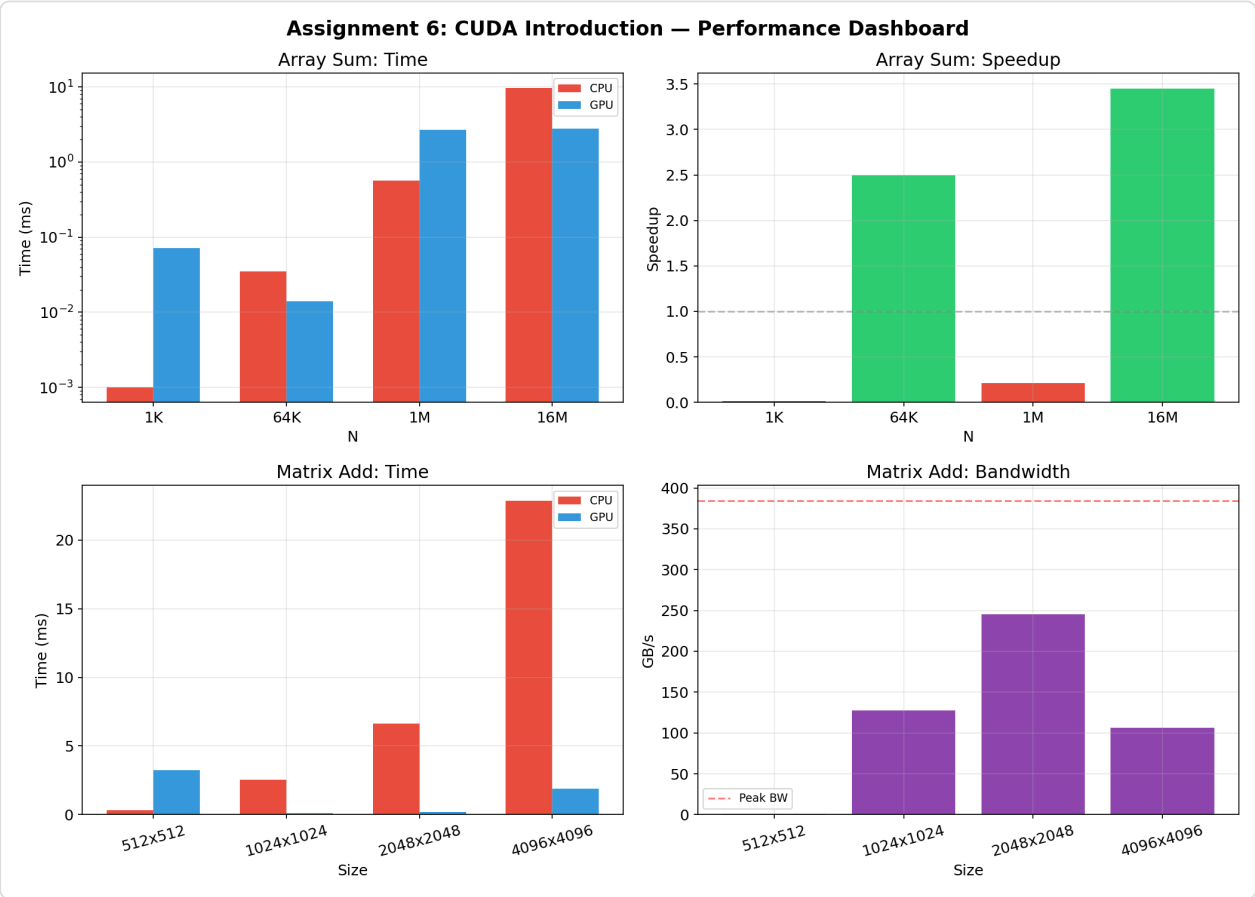
Metric	Formula	Example (1024x1024)
Floating-point ops	N x N = N^2	1,048,576
Global memory reads	2 x N^2 (read A and B)	2,097,152
Global memory writes	N^2 (write C)	1,048,576
Total bytes transferred	3 x N^2 x 4 bytes	12,582,912



Analysis

Peak speedup of **32.2x** at 2048x2048. At 512x512, the matrix is too small — kernel launch overhead exceeds compute time. At 4096x4096, the GPU achieves only 106.2 GB/s (28% of theoretical 384 GB/s) due to integer operations being less optimized than float, and memory access patterns not fully coalesced at larger sizes. The 2048x2048 sweet spot achieves 245.1 GB/s (64% of peak), showing that matrix addition is entirely **memory-bound** — the single addition per element gives an arithmetic intensity of only 0.083 FLOP/byte.

Performance Dashboard



What I Learned

1. **Kernel launch overhead is significant**: For small problems (N < 64K), the ~5-10 us kernel launch cost exceeds computation time. GPU parallelism only pays off when there's enough work to amortize this overhead.

2. **Matrix addition is purely memory-bound**: With arithmetic intensity of 0.083 FLOP/byte (1 add per 12 bytes transferred), performance is entirely limited by memory bandwidth, not compute throughput.

3. **2D grid/block mapping is natural for matrices**: Using `dim3` for both grid and block dimensions maps cleanly to row/col indexing, making the code intuitive and ensuring coalesced memory access along rows.

4. **Parallel reduction requires multi-pass thinking**: Unlike element-wise operations, reduction needs careful synchronization (`__syncthreads`) and multiple kernel launches since blocks cannot synchronize with each other.

5. **Device query reveals hardware constraints**: Understanding max threads/block (1024), warp size (32), and shared memory limits (48 KB) is essential for writing efficient kernels. Exceeding these limits causes silent launch failures.

Compilation & Execution

```
# Build everything
make all

# Run individual parts
make run-a    # Device Query
make run-b    # Array Sum
make run-c    # Matrix Addition

# Run all
make run-exercises

# Clean
make clean
```